

A Mostly Harmless Introduction to Claude Code, Codex and Friends

For Empirical Researchers

Daniel Sánchez Pazmiño

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Welcome

- Workshops for Ukraine
- Who this talk is for: empirical researchers, not software engineers
- What we will and will not cover

About Me

Daniel Sánchez, MA

Consultant, Baker Tilly Ecuador — AI Enablement Expert

Economist specialized in automation; expert in applied AI and analytics for decision-making. Combines data analysis, visualization, finance, and audit.

Education

- MA in Economics — Simon Fraser University (Canada)
- BA in Finance — Universidad San Francisco de Quito (Ecuador)
- BA in Economics — Universidad San Francisco de Quito (Ecuador)

Areas of Expertise

- Applied analytics for decision-making
- Power Query and data-flow automation for accounting and audit
- AI integrations for analysis and reporting
- Visualization and communication of results

Agenda

- 1 Introduction: why agentic AI
- 2 What's out there: the landscape of tools
- 3 How to use agentic AI: mechanics and cost models
- 4 Quick overview: best model for common users
- 5 Usages (the core, hands-on section)
- 6 Cybersecurity: do's and don'ts
- 7 Multi-agent (tentative)

Introduction: Why Agentic AI

- From autocomplete to agents: what changed
- Why this matters specifically for empirical research workflows
- Agents can read, edit, and run — not just suggest text

The Landscape: What's Out There

- Terminal / CLI agents: Claude Code, Codex, and friends
- Chat-based assistants vs. agentic tools
- IDE-integrated agents
- What researchers are actually adopting right now

Mechanics and Cost Models

- How an agentic session actually works under the hood
- Context windows, tool calls, and iteration loops
- Subscription vs. API/token-based pricing
- Ballpark costs for a typical research workflow

Quick Overview: Best Model for Common Users

- What “best” depends on: task type, budget, workflow
- Rules of thumb for picking a model/tool as a non-specialist
- When the default choice is good enough

Usages

The core, hands-on section.

1. Code Review

- Catching bugs and silent errors in statistical/data code
- Reviewing before merging or before submitting to a coauthor

2. Literature Review

- Search, summarize, and synthesize across papers
- Where agents help — and where they still hallucinate citations

3. Writing Review

- Drafting and editing papers, referee reports, slides
- Keeping the researcher's voice and claims intact

4. Skills and Agents from Modern Economists

- Reusable prompts, skills, and workflows built by other researchers
- Examples worth adapting

Cybersecurity: Do's and Don'ts

Do

- Use vetted, org-approved tools and check for a data processing agreement
- Keep credentials out of prompts — use secret managers/env vars, never paste tokens or passwords into chat
- Scope MCP/data connections to least privilege (read-only where possible)
- Know what data leaves your machine before granting file or network access

Don't

- Don't paste PII, credentials, or restricted admin microdata into a public chat interface
- Don't grant blanket filesystem or internet access without understanding what the agent can reach
- Don't treat “the agent says it's fine” as a compliance answer
- Don't skip your institution's confidentiality and data-handling

Multi-Agent

(Tentative — first thing to cut if running long)

- Running multiple agents in parallel or in a pipeline
- When one agent isn't enough: division of labor, orchestration
- Practical caveats: cost, coordination, verification overhead

Thank You

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